

The Edge Negotiator

Verified-Source Cross-Junction Coordination for SLM-Driven Traffic Signal Control

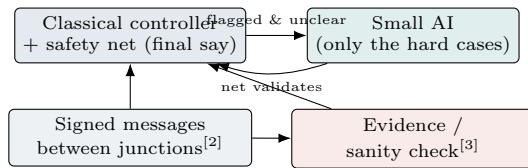
Progress report · UCL MSc Systems Engineering for IoT (25153651) · Supervisors: Dr A. Delibasi (UCL), L. Stott (Microsoft) · 2026-06-20

1. The project, and where it is headed

Picture a row of traffic junctions. Each one is run by a small AI model (**Phi-4-mini**) on the junction itself, through **Microsoft Foundry Local**, never retrained. The junctions tell each other what they are about to do, over a channel that is digitally signed and sanity-checked. Underneath sits a fast classical controller (*MaxPressure*^[1]). It is the default, and it is the safety floor.

The goal is simple to state. Coordinate well in normal traffic, and **fail gracefully when a neighbour lies**. If a junction's messages are tampered with, or the junction itself is hijacked, the system should notice and fall back safely instead of believing it. A deterministic safety net always makes the clear-cut safety calls: clear an ambulance it can actually see, and refuse a claim it cannot verify. Safety never depends on the AI. **The AI handles only the hard cases a fixed rule cannot settle**. When an alert is flagged but the evidence is mixed, it weighs what the neighbours report and judges: real incident, or faked grab? Whether that on-device reasoning beats a well-tuned fixed rule is the central question we measure. We will report the answer honestly, even if the rule turns out to be enough.

2. How it works



Normally the classical controller runs. The safety net has the final say on every safety call: clear a real ambulance, refuse a claim it cannot verify. When an alert is flagged but unclear, the AI weighs neighbour reports to judge real versus fake. Every message is signed and checked first.

4. What we have done

- **Built and tested:** signed messaging between junctions, an approved-senders list, the classical controller with its safety net, a check that flags neighbour reports that do not match the real traffic, and a four-junction simulated road.
- **Emergency handling:** a junction clears an ambulance only when it has real proof the vehicle exists. A valid signature is not enough. So a faked emergency is refused.
- **Live attack-and-defence demo:** a hijacked junction sends a signed but fake “ambulance coming” alert. It is refused. A real ambulance is cleared at every junction. We compare against a

5. Why it is new

No one has tested a setup like this: many junctions, coordinated by AI, where a junction's messages can be *faked by an insider that still holds a valid key*. The parts exist separately. The combination does not. Closest work: **CoLLMLight**^[5] (AI junctions that coordinate, but trust each other completely); **VLMLight**^[6] (fast-classical plus slow-AI, one junction, very large models, no trust); **LA-Light**^[7] (AI for rare events, cloud-based, no malicious faults); **Keijzer 2021**^[8] (detects coordination attacks, but only in theory). Our difference: a small on-device AI, a *measured* picture of how detection holds up as the attack grows, verifiable identity, and a tamper-proof log.

6. Next steps

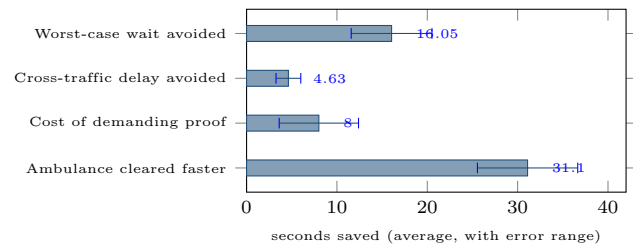
- **The central experiment:** on flagged, unclear cases, does the AI actually beat a well-tuned fixed rule? We measure this and report it either way.
- Bigger evaluation: more traffic levels and a real London (Lambeth) road, with many repeated runs and error ranges.
- Find the breaking point: grow the lie until detection fails, and

7. What we do and do not claim

We claim: trustworthy messaging between junctions; reliable detection of faked or faulty reports, with a measured reliability curve; a guaranteed safety floor; and a fair attack-and-defence test. **We do not claim:** new cryptography; that the AI beats the classical controller on ordinary traffic; that the AI already improves the hard-case decisions (we measure that next); defence against two or more junctions working together, or a stolen master key; or that simulator results carry straight over to real hardware (future work).

References. [1] Varaiya, MaxPressure control, *Transp. Res. C* 36 (2013). [2] Bernstein et al., Ed25519 signatures (2012); approved-sender registry. [3] Lighthill & Whitham, *Proc. R. Soc. A* 229 (1955); Richards, *Oper. Res.* 4 (1956): the traffic conservation law behind the sanity check. [4] Lopez et al., Microscopic Traffic Simulation using SUMO, *IEEE ITSC* (2018). [5] Yuan et al., CoLLMLight, arXiv:2503.11739 (2025). [6] Wang et al., VLMLight, arXiv:2505.19486 (2025). [7] Wang et al., LA-Light, arXiv:2403.08337 (2024). [8] Keijzer et al., Detection of Cyber-Attacks in Collaborative Intersection Control, *ECC* (2021).

3. Main result (measured over 30 runs)



Our system against two baselines, on the same traffic (SUMO^[4] simulator). All four results are clearly above zero.

trust-everything system and the plain classical controller.

- **Evidence (30 runs):** emergency handling clears the ambulance **31 s faster** than having none. The defence blocks an attack that would otherwise add **16 s** to the worst side-street wait. Demanding proof costs **8 s** versus blindly trusting; that cost is deliberate.
- **Independent check:** an automated audit of our written spec found *no made-up sources*. A few over-stated citations were fixed. Every result can be reproduced from its settings.